

Highland (Mountain / Elevation, Airspace, Defensive Warfare) Map Design

Executive Summary

Highland is a 3D-rendered mountain battlefield built for atmospheric combat, airspace control, and layered defensive warfare. The terrain features steep ridges, dense cliffs, fortified passes, elevated artillery platforms, and narrow mountain roads. Airspace above is a major color domain. VTOL craft and high-altitude sensors dominate. The map simulates a Coldwar/Postwar's autonomous elevation base, air-ground coordination, complex relay networks, and battles under harsh mountain constraints. It is 88.7km (55mi) by 11.5km (7.1mi) in size, with a 100m (328ft) grid. The map is divided into 4 quadrants, each with a 100m (328ft) grid. The map is divided into 4 quadrants, each with a 100m (328ft) grid. The map is divided into 4 quadrants, each with a 100m (328ft) grid.

Design Goals & Gameplay

Elevation-Driven Combat

Highland prioritizes verticality. High ground positions dominate sightlines, artillery arcs, and sensor coverage. Controlling ridges is essential.

Airspace & Aerial Coordination

Air units are treated as separate entities, often low corridors, and off-limits for ground units. Ground units rely on aerial recon for terrain and movement.

Autonomy & Elevation Logic

Autonomous AI entities (AIOP units), mobile, respond to threats during enemy incursions, and uses GPS cover to break line of sight.

Sensor & Information Warfare

This map uses fog, particle, and elevation gradients to obscure vision. Enemy bases are peaks, not easy to identify. Destroying relay creates massive blind zones.

Logistics & Resources

Mountain passes, valleys, and fortified bases serve as supply routes. Supply convoys must navigate narrow roads and avoid ambush zones.

Deployment & Progress

A base deployment is limited to 10,000 points/force players to choose between heavy artillery, air support, or mobile mountain units. Base deployment from the southwest valley base, red deployment from the northeast peak fortress.

Spatial Constraints (Nreal AR)

Visual clarity: 100m (328ft) grid resolution. Optimal viewing distance: 10-15m.

Playstyle Area & Scaling

100m (328ft) grid resolution. Optimal viewing distance: 10-15m.

Tracking & Occlusion

Inside-out tracking allows horizontal planes. Depth-mesh (if available) provides limited occlusion. Virtual objects render near real-world color, ensure minimal map-world exclusion.

Layout Overview

Coordinate System: 1000x1000 grid (1 unit = 1 m). Origin (0,0) is southwest corner.

Key Locations

Blue HQ (Valley Base) - (0,0) to (500,500) in Mountain-Flat base with artillery and recon drones.

Red HQ (Peak Fortress) - (500,500) to (1000,1000) in Mountain-Flat base with recon and AA emplacements.

Central Mountain Ridge - (500,0) to (500,1000) in Mountain-Flat base with recon and AA emplacements.

Mountain Base - (0,500) to (500,1000) in Mountain-Flat base with recon and AA emplacements.

Mountain Ridge - (500,0) to (500,1000) in Mountain-Flat base with recon and AA emplacements.

Red Base - (500,0) to (500,1000) in Mountain-Flat base with recon and AA emplacements.

Lower Valley Trail - (0,0) to (500,500) in Mountain-Flat base with recon and AA emplacements.

Peak A - (500,0) to (500,1000) in Mountain-Flat base with recon and AA emplacements.

Peak B - (500,0) to (500,1000) in Mountain-Flat base with recon and AA emplacements.

Nearby Forward Outpost - (500,0) to (500,1000) in Mountain-Flat base with recon and AA emplacements.

Chokepoints & Sightlines

Ridge Pass

Primary vertical, ideal for defensive artillery and ambushes.

Cliffside Barker Network

Primary vertical, ideal for defensive artillery and ambushes.

Highland Plateau

Primary vertical, ideal for defensive artillery and ambushes.

Lower Valley Trail

Primary vertical, ideal for defensive artillery and ambushes.

Sightlines

Peak A provides long-range sensor coverage (1000 m). Fog patches and elevation gradients create blind zones.

Deployment Zones

Blue Deployment: 100 valley base + recon units.

Red Deployment: 100 peak fortress + AA emplacements.

Nearby Forward Outpost: 1000 m (328 ft) for drone base.

Assets & Environment

Terrain

Cliff, ridge, plateau, valley, rock formation.

Vegetation

Scrub, grass, shrubs, forest, pine, oak.

Buildings & Props

Military: Barracks, HQ, gun, radar station, helipad, fuel tank, generator, pipeline, control tower. Logistics: Trucks, fuel drums, drones.

Effects

Fog, smoke, gas, dust, snow/freeze (optional), rainfall/sleet.

Audio

Wind, distant artillery, radio chatter, aircraft movement.

LOD & Budget

Asset List

Asset Name Count Triangles Total Size

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